IT PROF EL 4 - Advanced System Integration and Architecture

Semester Project, System Integration Documentation

INTEGRATING APPLICATIONS AND INTEGRATING THE ENTERPRISE

Submission Form

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Course/Section: BSIT / IT PROF EL 4 80107	**Date:** October 7, 2026

Chosen App or Organization (semester project):	**Student Lost & Found App**

Part 1. Integration Map and Legacy Check

1A. Integration map (list at least 4 connections between systems)

From (system)	To (system)	Data that moves	How it moves today (direct link, file, API, or by hand)
Student web/mobile app	Authentication system	Login credentials, user role	API (HTTPS)
Lost Item Reporting Module	Lost Item Database	Item name, description, location, date, status	Direct link
Lost Item Reporting Module	Image Storage	Uploaded item photos	Direct link (file upload)
Lost Item Database	Admin Review Dashboard	Submitted reports and claim requests	Direct link
School security office logbook	Lost Item Database	Items physically turned in	By hand

Which connections are point-to-point, and why can this become hard to manage as more systems are added? (2 to 3 sentences)

The links between the reporting module, the database, the image storage, and the admin dashboard are point-to-point because each pair is connected directly. As more systems are added (notifications, a student records system, a mobile app), the number of direct links grows quickly. A change in one system can then break several others, and every link must be maintained separately.

1B. Legacy check

Field	Entry
Older system, tool, or process the app still depends on	The paper logbook and physical lost-and-found area at the school security office, plus informal posts in group chats and social media

Why it is still used	Found items are physically turned in and stored there, and staff and students are already used to this process.
Main challenge in connecting it	Its records are handwritten, unstructured, and inconsistent, so the app cannot read or search them.
Approach to connect it (adapter or wrapper, middleware, staged replacement, data synchronization, or encapsulation)	Staged replacement
Why this approach fits	Staff keep the logbook while the app runs alongside it. Each turned-in item is also encoded in the app, and the logbook is retired once staff are comfortable. This avoids disrupting the office or losing records.

Part 2. Enterprise Integration: Shared Data

Shared data item (one data item used by two or more systems):	**Lost Item Report**

Shared data format (at least 5 fields)

Field name	Meaning	Example value
itemId	Unique ID of the report	LF-2026-0042
itemName	Short name of the item	Black umbrella
location	Where the item was lost or found	Library, 2nd floor
dateReported	Date the report was made (ISO 8601)	2026-09-15
status	Current state of the report	Lost / Found / Claimed
imageUrl	Link to the stored photo of the item	/images/LF-2026-0042.jpg

Systems that will use this shared format:	Lost Item Reporting Module, Search Module, Claim Module, Admin Review Dashboard, and Notification Service

Why is one shared format better than each system using its own format? (2 to 3 sentences)

With one shared format, every module reads and writes the same fields, so no conversion between formats is needed. This reduces errors and mismatched data, such as "location" in one system and "place" in another. It also makes adding new systems faster, since they only need to follow the one format.

Part 3. Integration Strategy Choice

Field	Entry
Chosen strategy (point-to-point, API-led or service-oriented, message or event-driven, or middleware/ESB)	API-led (service-oriented)
One advantage for this app	Each function (search, report, claim) is exposed as a reusable API, so a mobile app or notification feature can be added later without rebuilding existing parts.
One disadvantage for this app	The team must design, secure, document, and maintain the APIs, which takes extra effort for a small project.
Why this strategy fits this app (2 to 3 sentences)	The app is made of separate functions that all work with the same item data. An API-led approach lets each module use that data through a clear, secure interface instead of direct links, and it fits the shared Lost Item Report format.
One strategy not chosen, and why	Middleware/ESB. It is too heavy and costly for a small campus app with only a few systems, and the team does not need its advanced routing and transformation features.

Part 4. Service Entry and Best Practices

4A. Service entry (one specific service, not the whole app)

Field	Entry
Service name	Lost Item Search Service
What it does	Finds posted lost or found items that match a student's search
Who uses it	Students (to search), the Claim Module, and administrators
Input (what it needs)	Keyword, category, location, date range, and the user's access token
Output (what it returns)	A list of matching items (itemId, itemName, location, dateReported, status, imageUrl). Personal contact details are not included.
Data format (XML or JSON) and why	JSON. It is lightweight, easy to read, and works well with web and mobile apps.
Who is responsible for it	The project team (Chinley A. Suan and John Michael R. Cabug) for development and maintenance. The assigned school administrator oversees the data and reports.
Security control (for example, an API key)	API key/access token required, HTTPS, and role-based access

Version (start at v1)	v1 (/api/v1/items/search)

4B. Best practices applied (2 XML or data format practices and 2 web service practices)

Type	Practice chosen	How it applies to this service (one sentence)
XML or data format practice 1	Use clear, consistent field names	Search results always use the same names (itemName, dateReported, status), so every module reads them the same way.
XML or data format practice 2	Use a standard date format (ISO 8601) and validate data against a defined schema	Dates are returned as 2026-09-15, and requests with missing or wrongly formatted fields are rejected.
Web service practice 1	Version the service	The /v1 in the URL lets the team update the service later without breaking modules that use the current version.
Web service practice 2	Return clear error messages and proper HTTP status codes	An invalid token returns 401, and a search with no matches returns an empty list with a clear message.